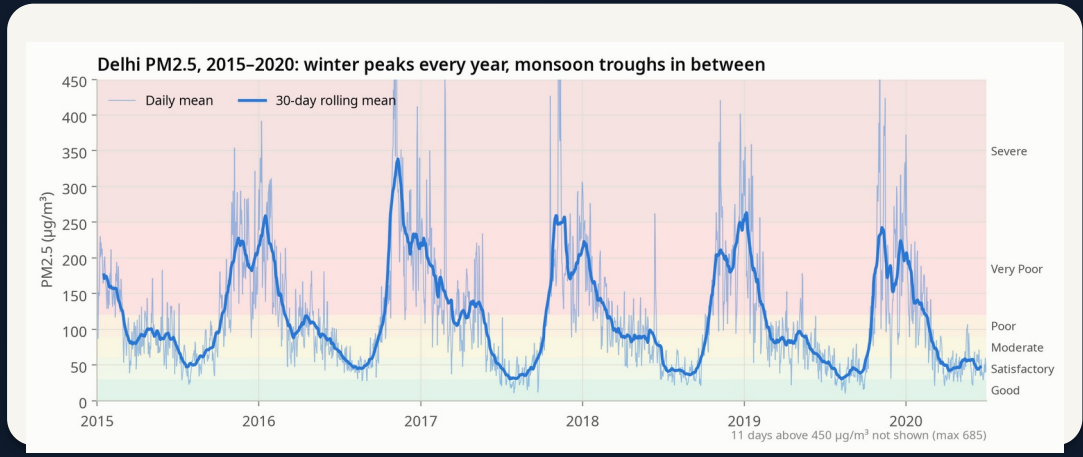
TIME SERIES ANALYSIS · PROGRESS UPDATE

Forecasting Delhi PM2.5 with SARIMAX, VAR and Granger causality



Goal: a 12-month AQI forecast with 95% intervals for health advisories, and evidence on whether wind and humidity drive PM2.5.

2,009

daily records, Jan 2015 to Jul 2020

5 series

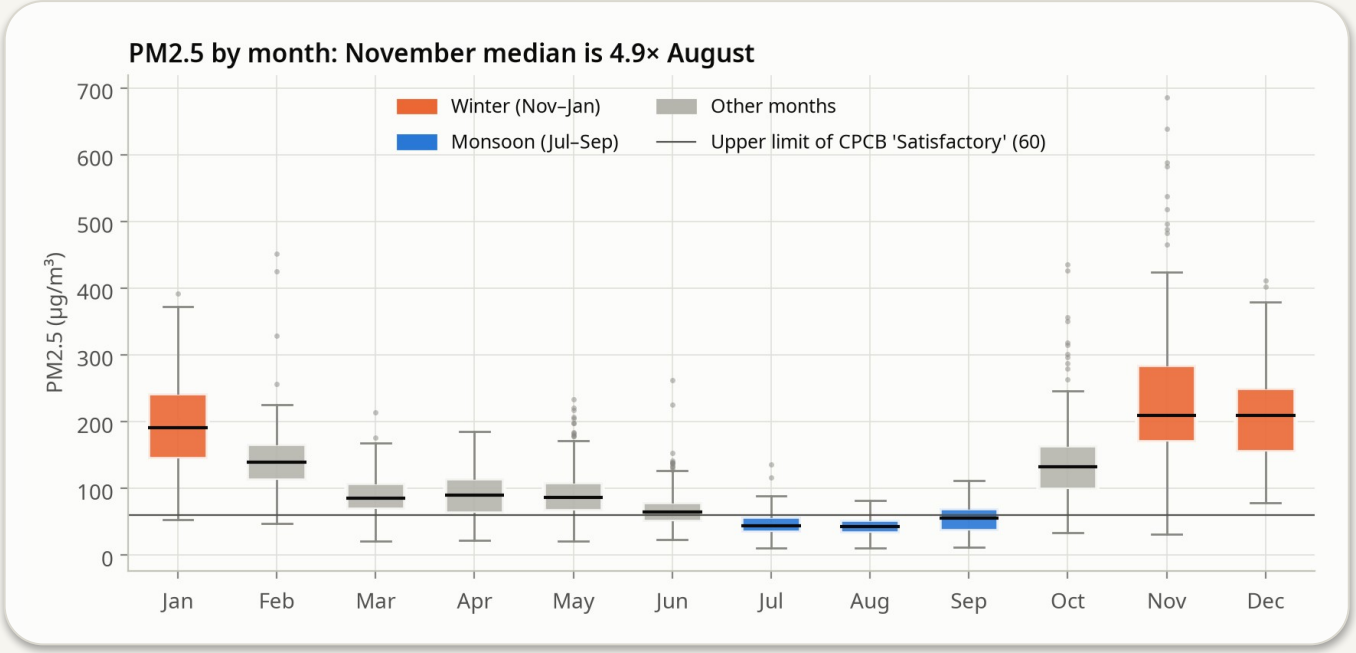
PM2.5, NO_2, SO_2 (CPCB); wind, humidity (ERA5)

52%

of days rated Poor, Very Poor or Severe

Data: CPCB via Kaggle, Open-Meteo ERA5

Seasonality dominates, and the series is stationary



Winter smog (Nov–Jan) and monsoon clearing (Jul–Sep) repeat every year, so a seasonal ARIMA with period 52 weeks is justified.

4.4×

Winter mean 213 vs monsoon 48 µg/m³

0.89

STL seasonal strength; 1 means pure seasonality

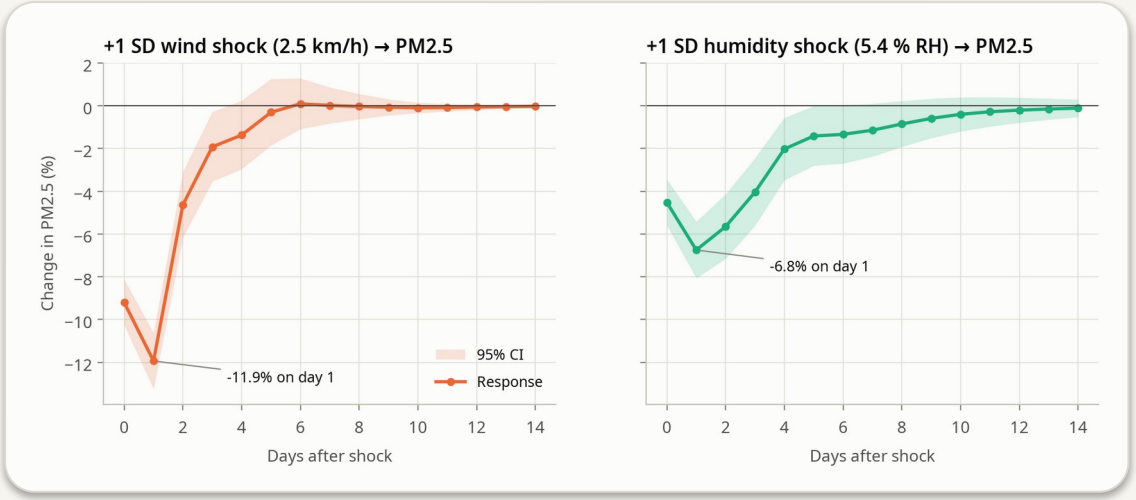
Stationary

Log PM2.5 passes ADF and KPSS; D = 0 vs 1 is decided in SARIMAX

VAR and Granger: weather leads PM2.5, not the reverse

Direction	p-value	Verdict
Wind → PM2.5	1.7e-29	Causes
Humidity → PM2.5	1.1e-11	Causes
PM2.5 → Wind	0.13	No evidence
PM2.5 → Humidity	0.61	No evidence

VAR(4) on deseasonalised daily data, 2015–2019. Stable (max $|\lambda| = 0.81$) with white residuals (Portmanteau p = 0.17). Same result with the BIC lag and with first differences.



- +1 SD wind (2.5 km/h): PM2.5 down ≈12% next day
- +1 SD humidity (5.4% RH): PM2.5 down ≈7%
- Weather explains ≈30% of 14-day forecast error

What is done, and the road to the forecast

Done so far

- Cleaned and merged 2,009 days of CPCB and ERA5 data; 2 missing PM2.5 days interpolated
- Showed a strong annual cycle and a stationary log PM2.5 series
- Showed wind and humidity Granger-cause PM2.5, one way only and robust to the model setup

Caveat: Granger shows prediction, not physical cause. Mixing height and stubble burning are not measured.

Next

1. SARIMAX

Weekly log PM2.5, period 52, wind and humidity as exogenous inputs; orders by AIC/BIC, Ljung-Box residual check

2. Baselines

Linear Regression and LSTM against seasonal naive; compare test MAE on 2019

3. 12-month forecast

Refit on all data; 52 weeks with 95% intervals and a monthly AQI advisory table

Plan: train 2015–2018, test 2019, lockdown 2020 as a case study